Dynamics-Aware Gradient Clipping: Protective Feedback for Unstable Training

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Abstract

Gradient clipping prevents destructive updates but a fixed threshold must be retuned when the optimizer, learning rate, or model changes. We introduce Dynamics-Aware Gradient Clipping (DAGC), a lightweight controller that changes a global clipping threshold from the recent clipping exposure and gradient alignment. An initial controller (DAGC-V1) drove the threshold in one direction only: because its exposure term is non-negative for any exposure level, the exposure alone could relax the threshold but could not tighten it. DAGC-V2 makes the same term bidirectional, so realized exposure above a reference level lowers the threshold. We select V2 on separate development seeds and evaluate it with eight paired, held-out seeds per method. On unstable FashionMNIST training with SGD, V2 improves over calibrated fixed clipping by 0.82–2.18 percentage points across two learning rates, and by 1.53 points on a different MLP architecture. On MNIST with a CNN it improves by 1.16 points. It also exceeds adaptive gradient clipping (AGC) in each condition. A sweep of the reference level from 0.1 to 0.7 leaves accuracy unchanged to within 0.7 points while realized exposure stays at the ceiling throughout: what the correction buys is a threshold that can fall, not the tracking of any particular exposure level. These are diagnostic-scale MNIST-family experiments. Additional paired controls show that exposure removal costs 5.47 points, while removing alignment costs only 0.16 points and a calibration-derived open-loop schedule nearly matches V2. Thus the evidence supports a protective threshold profile without establishing an additional benefit from online directional feedback. The results establish a controlled signal rather than broad large-scale generalization.

1. Introduction

Gradient clipping is a standard safeguard for exploding or destructive updates (Pascanu et al., 2013). In its usual global-norm form, however, the threshold is constant throughout training. A threshold that is useful during a transient spike can over-constrain later updates, whereas a permissive threshold can fail in an unstable phase. This creates a per-setting tuning burden and makes clipping behaviour difficult to transfer across learning rates and architectures.

We study clipping as a feedback-control problem. DAGC maintains a scalar global threshold and observes two signals already available in a training loop: whether clipping was active and whether consecutive gradients agree in direction. Its first version had a one-sided exposure response. Because $E_t \in [0, 1]$, the V1 exposure term $r(1 - E_t)$ is non-negative at every exposure level, so the exposure alone could only increase the threshold: even at maximal exposure it merely slowed relaxation. The alignment term could still tighten the threshold, but persistent clipping alone could not trigger that response. DAGC-V2 replaces the term with $r(\tau - E_t)$, which changes sign, so excess exposure actively lowers the threshold.

Our evidence is deliberately scoped, and one of our results revised our own account of the method. We first use diagnostic runs to locate an unstable learning-rate band and identify the V1 failure mode. We freeze V2 after a separate three-seed development comparison, calibrate only the fixed baseline on separate seeds in each new condition, and report eight paired held-out seeds per method. The development comparison also swept the reference level τ; since that sweep leaves results unchanged and realized exposure stays near 1.0, we report V2 as a bidirectional controller rather than as a controller that tracks a target exposure, and we present the sweep as evidence for that reading. The contributions are: (i) bidirectional exposure feedback for global gradient clipping, with an explicit account of what the correction does and does not do; (ii) a reproducible calibration/evaluation protocol that separates controller selection, baseline tuning, and evaluation; and (iii) controlled results across learning rates, architectures, and MNIST-family datasets, including negative controls where no clipping diverges.

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2. Related Work

Fixed and adaptive clipping. Global-norm clipping rescales a gradient whose norm exceeds a constant threshold and is the standard remedy for exploding gradients (Pascanu et al., 2013); it is retained in most large-scale training recipes (Chowdhery et al., 2023). Its weakness is that the threshold must be re-tuned whenever the model, batch size, or learning rate changes. A first line of work replaces the constant with a statistic of the current gradient scale. AutoClip takes a percentile of the gradient-norm history (Seetharaman et al., 2020); adaptive gradient clipping (AGC) bounds the ratio of gradient norm to parameter norm (Brock et al., 2021); Clippy combines relative and absolute thresholds and switches between them during training (Tang et al., 2023); ZClip detects spikes by the z-score of the gradient norm under an exponential moving average (Kumar et al., 2025); and per-tensor rules such as AdaGC adapt a separate threshold per parameter tensor (Wang et al., 2025). DAGC belongs to this family but keeps a single global threshold, so it does not compete with per-tensor granularity. Its distinguishing feature is the control signal: these methods infer the threshold from the magnitude distribution of recent gradients, whereas DAGC infers it from whether clipping was actually binding and whether consecutive gradients agree in direction, which makes the controller responsive to its own intervention rather than to the norm alone. Event-frequency feedback itself is not new, however.

Quantile and clipping-rate feedback. Andrew et al. (2021) estimate a target quantile of client-update norms online in private federated learning. Their geometric threshold update can be written, using clipped fraction E and desired clipped fraction τ, as $c^+ = c\exp(\eta(E-\tau))$: excess clipping raises the threshold. Merad & Gaïffas (2024) study gradient quantile clipping for robust stochastic optimization, including heavy-tailed samples and contamination. Shulgin & Richtárik (2024) analyze quantile clipping and its private extension, identifying clipping bias and schedules that mitigate it. These are close precedents for adaptive thresholds and rate-based feedback. DAGC-V2 instead uses temporal exposure and gradient reversal as protection signals: its exposure term has the opposite sign, so persistent clipping lowers the threshold. It does not estimate or track a norm quantile. Accordingly, our contribution concerns the tested protective controller, rather than the first use of clipping-rate feedback or multiplicative threshold adaptation.

Theory of clipping and step size. Clipping is justified when smoothness is weak, because it restores a usable effective step size (Zhang et al., 2020a), and its convergence properties under heavy-tailed noise are now well characterised, including the bias introduced by biased clipping estimators (Zhang et al., 2020b; Koloskova et al., 2023). These analyses do not establish convergence of our controller; quantile-clipping analyses (Merad & Gaïffas, 2024; Shulgin & Richtárik, 2024) likewise cannot be transferred directly to DAGC-V2 because its threshold dynamics differ.

Mechanisms of instability. Instability is commonly described through the margin between step size and local curvature: descent becomes unstable once the learning rate exceeds roughly $2/\lambda_{\max}$ of the loss Hessian (Cohen et al., 2021; Gilmer et al., 2022), and this account is consistent with loss-spike reports from large-scale pre-training (Chowdhery et al., 2023). Sharpness-aware objectives act on the same quantity from the opposite direction, penalising sharp minima directly (Foret et al., 2021). A complementary view attributes large updates to the heavy-tailed rather than Gaussian nature of stochastic gradient noise (Şimşekli et al., 2019), which is the regime our unstable band occupies. We estimate no curvature and do not test these mechanisms; we use them only to motivate why a learning-rate band with genuine divergence is the right place to look.

Optimizer- and architecture-level stabilisation. Not all stability work intervenes on the gradient. Update-level clipping and low-precision stabilisers such as LAMB and StableAdamW act through the optimizer (You et al., 2020; Wortsman et al., 2023); learning-rate warmup and scaling rules control the early phase in which curvature is steepest (Goyal et al., 2017; McCandlish et al., 2018); momentum and careful initialisation shape the same dynamics from the start (Sutskever et al., 2013; Glorot & Bengio, 2010), as does the choice of optimiser itself (Kingma & Ba, 2015; Krizhevsky et al., 2012); and normalisation layers are the dominant architectural remedy, whether or not their benefit derives from the mechanism originally proposed for them (Ioffe & Szegedy, 2015; Santurkar et al., 2018; Ba et al., 2016; Zhang & Sennrich, 2019). Weight decay is likewise used for stability rather than for its classical regularisation effect (D'Angelo et al., 2024). DAGC is orthogonal to all of these and, in principle, composable with them; our experiments therefore hold the optimiser, initialisation, and architecture fixed and vary only the clipping rule.

Measurement practice. Benchmark conclusions depend strongly on variation from data sampling, initialisation, and hyperparameter choice, and underestimating that variance is a common reason a reported improvement fails to replicate (Bouthillier et al., 2021). We therefore make the seed, rather than the training iteration, the unit of comparison, report paired differences, and treat divergence as a primary outcome. Restricting instability studies to small models at high learning rates is an established proxy for large-scale behaviour rather than a concession (Wortsman et al., 2024), although the scale limits what can be claimed and we flag it wherever it applies.

Algorithm 1 DAGC-V2 at training step t

1: Compute $\|g_t\|_2$ and a_t; clip g_t at c_t.
2: Set $e_t \leftarrow \mathbb{1}[\|g_t\|_2 > c_t]$.
3: Update $E_t \leftarrow \beta E_{t-1} + (1-\beta)e_t$ and A_t analogously.
4: $u_t \leftarrow r(\tau - E_t) - w\max(0, -A_t)$.
5: $c_{t+1} \leftarrow \Pi_{[c_{\min}, c_{\max}]}[c_t \exp(\gamma \tanh(u_t))]$.
6: Apply the optimizer update using the clipped gradient.

3. Method

Let g_t be the gradient and $c_t > 0$ the current threshold. We apply global norm clipping,

$$\tilde{g}_t = \min\{1, c_t/(\|g_t\|_2 + \epsilon)\}g_t. \tag{1}$$

Let $e_t = \mathbb{1}[\|g_t\|_2 > c_t]$ denote clipping exposure and let $a_t = \langle g_t, g_{t-1}\rangle/(\|g_t\|_2\|g_{t-1}\|_2 + \epsilon)$ denote consecutive-gradient alignment. We maintain exponential moving averages $E_t = \beta E_{t-1} + (1-\beta)e_t$ and A_t of these signals. DAGC-V2 updates

$$c_{t+1} = \Pi_{[c_{\min}, c_{\max}]}\left[c_t \exp\left(\gamma \tanh\left(r(\tau - E_t) - w\max(0, -A_t)\right)\right)\right], \tag{2}$$

where τ is a reference exposure level. The exposure term $r(\tau - E_t)$ is negative when $E_t > \tau$ and positive when $E_t < \tau$, so the controller can both raise and lower the threshold, and τ sets the sign changeover rather than a level the controller is expected to reach. The alignment term conservatively tightens the threshold after reversal-like dynamics. The projection bounds are scaled by a rolling median of recent gradient norms. The method adds scalar moving averages, one gradient norm, and one gradient dot product per update.

DAGC-V1 is the special case obtained by replacing $r(\tau - E_t)$ with $r(1 - E_t)$, which amounts to setting the reference level to 1. Since $E_t \in [0, 1]$ the V1 exposure term is then non-negative for every reachable exposure. Thus exposure alone cannot tighten the threshold, although the negative-alignment term and moving bounds can still decrease it. V2 removes that constraint on the exposure response. The modification is therefore a directly testable one-sign change rather than an unrelated new algorithm. As we report below, the change is what matters and the particular value of τ is not, because realized exposure under V2 settles near 1.0 and so the term stays active at almost every step regardless of τ.

The order in Algorithm 1 matters: clipping at c_t determines the exposure observation, and the controller affects the next update. This avoids a circular within-step threshold choice. In our implementation, the controller alignment is computed from consecutive full gradients. Fixed random projections are used only for auxiliary diagnostics; retaining the previous full gradient requires storage proportional to the parameter count.

4. Why Protective Feedback Can Help

A bounded intervention. Ignoring the numerical epsilon, global clipping satisfies $\|\tilde{g}_t\| \le c_t$. For momentum SGD without weight decay, $v_t = \mu v_{t-1} + \tilde{g}_t$ and $x_{t+1} = x_t - \eta v_t$. If $c_j \le C$ over a window of k steps, the triangle inequality gives

$$\|v_t\| \le \mu^k \|v_{t-k}\| + C\frac{1-\mu^k}{1-\mu}. \tag{3}$$

Consequently, sustained smaller thresholds bound newly accumulated momentum; they do not instantly erase the existing buffer. This elementary bound explains a possible protection mechanism, not convergence of the full training process. With coupled weight decay, an additional parameter-dependent term enters the buffer and must be included in any full bound.

A minimal oscillation model. For $f(x) = \lambda x^2/2$, unmodified momentum SGD has recurrence $x_{t+1} = (1 + \mu - \eta\lambda)x_t - \mu x_{t-1}$. For $0 \le \mu < 1$, its linear stability condition is $0 < \eta\lambda < 2(1 + \mu)$. Without momentum, when $\eta\lambda > 2$, successive iterates reverse sign and grow in magnitude; successive gradients therefore have alignment -1. Clipping bounds the gradient contribution even when this linear stability condition fails. Direction reversal is thus a plausible warning signal in this model, but stochastic minibatches can also reverse direction during useful learning. Its incremental value in neural networks requires a dedicated V2 ablation.

Protection is different from rate tracking. For a fixed distribution of gradient norms, lowering c increases $\Pr(\|g\| > c)$. V2 therefore reinforces clipping after high exposure rather than providing negative feedback toward a target clipping rate. Its bounds limit threshold collapse, but frequent activation can drive it toward the lower bound. The observed near-saturated exposure is consistent with this mechanism. It also admits a competing explanation: a small threshold or its lower bound, rather than state-dependent feedback, may account for the accuracy gain.

Testable predictions and evidential limits. The protective interpretation predicts that sustained activation reduces the gradient contribution to subsequent updates, and that reversal feedback adds value when oscillations precede loss deterioration. Three discriminating comparisons distinguish the proposed mechanism: quantile/rate tracking and AutoClip; paired V2 variants removing exposure or alignment; and fixed or predetermined thresholds matched using calibration traces. We report these comparisons below. Time-resolved logs are saved, but temporal associations alone are not treated as evidence that reversal causes future deterioration.

5. Experimental Protocol

All experiments use SGD with momentum 0.9, weight decay 10^{-4}, batch size 256, fifteen epochs, and 5,000 training / 2,000 test examples. Each result is the mean over eight paired held-out seeds with a normal-approximation 95% confidence interval. Paired differences use the same data subset, initialization, and minibatch order for every policy within a seed.

We select V2 once on FashionMNIST with a small CNN at learning rate 0.8, using development seeds 200–202. The frozen setting is $c_0 = 0.5$, $\gamma = 0.05$, $\beta = 0.9$, $r = 0.3$, $w = 3.0$, and $\tau = 0.1$. The same development runs sweep $\tau \in \{0.1, 0.2, 0.35, 0.5, 0.7\}$ crossed with $\gamma \in \{0.05, 0.1\}$; we report that sweep in full below because it determines how the correction should be interpreted. Evaluation seeds are disjoint: 300–307 for FashionMNIST-CNN at 0.8, 308–315 for the same model at 0.4, 500–507 for FashionMNIST-MLP at 0.8, and 700–707 for MNIST-CNN at 0.8.

In every condition, the fixed global-norm threshold is selected only from three separate calibration seeds. The selected threshold is 0.25 for FashionMNIST with the CNN and 0.10 for FashionMNIST-MLP and MNIST-CNN. We compare no clipping, this calibrated fixed rule, AGC with its standard parameter-relative coefficient 0.01 (Brock et al., 2021), DAGC-V1, and frozen DAGC-V2. A run is flagged as divergent for nonfinite loss or gradient norm, loss above 100, or a loss exceeding $\max(50, 20\ell_{t-1} + 1)$.

Adaptive baselines at their published defaults. Because V2 and the adaptive rules surveyed above address the same tuning burden, we additionally run ZClip (Kumar et al., 2025) and AdaGC (Wang et al., 2025). Both are held at the values published by their authors and are not tuned, which we label explicitly so that their numbers describe out-of-the-box behaviour rather than the best each method can do. One adjustment is unavoidable and is applied to both equally: their published warm-up lengths (25 steps for ZClip, 100 for AdaGC) occupy most of a diagnostic run, so each is given a warm-up proportional to the run length. Everything else — the z-score threshold and smoothing factor for ZClip, the relative and absolute thresholds and EMA coefficient for AdaGC — is untouched. We also record AGC at the same published default rather than tuned, for the same reason.

Diagnostic studies. We retain two preliminary diagnostics to make controller development auditable. A five-seed ablation (A3) uses FashionMNIST-CNN, SGD at learning rate 0.05, five epochs, and the same subset sizes. It compares the original controller with no exposure feedback, no alignment feedback, a frozen threshold, and fixed global clipping. A second 160-run grid (A4) crosses MNIST and FashionMNIST, MLP and CNN, SGD and AdamW (Loshchilov & Hutter, 2019), and four policies at stable learning rates. A3 and A4 are not part of V2 model selection; they are behavioural and negative controls.

Implementation and reporting. All policies in a paired comparison receive the same seed-controlled data subset, model initialization, and data-loader generator. We use PyTorch and a single CUDA device. The code writes one row per seed and policy, including test accuracy, divergence, clipping frequency, clipping intensity, and mean loss. Confidence intervals are $\bar{x} \pm 1.96 s/\sqrt{n}$; training steps are not treated as independent replicates. Evaluation scripts skip completed seed-policy pairs, making interrupted runs resumable without changing the protocol.

6. Results

Table 1 and Figure 1 report the held-out results. V2 is the best clipped method in each condition. At FashionMNIST-CNN with learning rate 0.8, it improves over fixed clipping by 2.18 points (7/8 paired wins) and over AGC by 1.61 points (8/8 wins). At learning rate 0.4, the gain is smaller but has the same direction: 0.82 points over fixed clipping and 0.56 over AGC. This is useful because it checks that the result is not tied to the single learning rate used during V2 development.

The architecture and dataset checks preserve the pattern. On FashionMNIST with the MLP, V2 improves over fixed clipping and AGC by 1.53 and 1.84 points, respectively, winning all eight seed-wise comparisons. On MNIST with the CNN, the absolute accuracy is near saturation, but V2 still gains 1.16 points over fixed clipping and 1.55 over AGC, again with eight paired wins. No clipping is a meaningful negative control in these high-learning-rate settings: its divergence rate is 88%, 100%, 100%, and 75%, respectively, in the table order.

The two adaptive baselines are reported at their published defaults, and under this protocol neither is competitive in the unstable band: both diverge on most seeds while V2, fixed clipping, and AGC do not. We draw only a weak conclusion from this. It does not show that their mechanisms are ineffective — tuning their thresholds on calibration seeds is the matching comparison and we have not run it — and it would be wrong to read a default-hyperparameter result as a statement about ZClip or AdaGC as methods. What it does show is that the tuning burden DAGC is meant to remove is not removed for free by every published adaptive rule, and that a claim about adaptivity has to be supported by a calibration protocol of the kind used here (Bouthillier et al., 2021). A symmetric comparison with a per-method tuning budget is the natural next step and is listed among our limitations.

Table 1. Held-out evaluation (eight paired seeds per policy). Fixed thresholds are selected on disjoint calibration seeds; V2 is frozen before all held-out evaluations. Values are mean test accuracy with 95% confidence intervals, rounded half-up to two decimals. Δ is the paired V2 advantage in percentage points, computed from unrounded means.

Dataset / model	LR	Fixed	AGC	DAGC-V1	DAGC-V2	$\Delta_{\mathrm{V2-Fixed}}$
FashionMNIST / CNN	0.4	86.76 ± 0.40	87.02 ± 0.70	83.18 ± 0.81	$\mathbf{87.58 \pm 0.63}$	$+0.82$ (6/8)
FashionMNIST / CNN	0.8	84.38 ± 1.19	84.95 ± 0.63	78.35 ± 4.67	$\mathbf{86.56 \pm 0.61}$	$+2.18$ (7/8)
FashionMNIST / MLP	0.8	82.59 ± 0.83	82.29 ± 1.12	77.51 ± 1.99	$\mathbf{84.13 \pm 0.52}$	$+1.53$ (8/8)
MNIST / CNN	0.8	97.23 ± 0.43	96.84 ± 0.36	85.19 ± 7.41	$\mathbf{98.39 \pm 0.15}$	$+1.16$ (8/8)

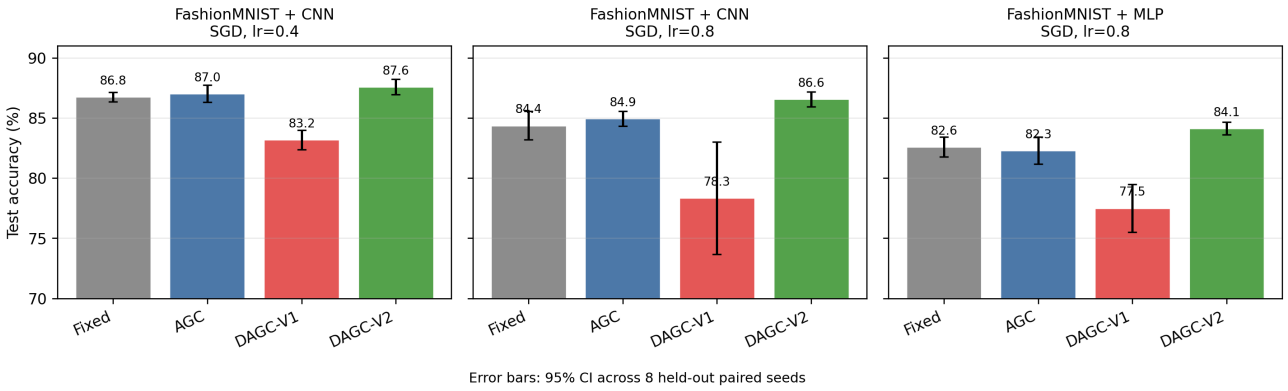


Figure 1. Held-out accuracy across the first three unstable-training conditions. Each bar summarizes eight paired evaluation seeds; error bars are 95% confidence intervals. The fixed baseline is separately calibrated in each condition, while DAGC-V2 remains frozen.

The V1 columns expose why the correction matters. V1 is lower than fixed clipping in all four conditions and has particularly large seed variation at FashionMNIST-CNN 0.8 and MNIST-CNN 0.8. This is consistent with the control analysis: since the V1 exposure term is non-negative at every exposure level, heavy clipping alone cannot make the threshold fall, which is consistent with insufficient protection where clipping is needed. Its mean realized clipping intensity at FashionMNIST-CNN 0.8 is 0.349, against 0.795 for V2 (0.505 versus 0.810 on the MLP and 0.340 versus 0.433 on MNIST-CNN), which is the predicted symptom.

What the correction buys, and what it does not. Table 2 and Figure 2 report the development sweep. The two candidate explanations make different predictions: if V2 worked by driving realized exposure towards τ, accuracy should depend on τ; if it worked by letting the threshold fall, accuracy should be insensitive to τ and realized exposure should saturate. The data support the second reading. Over a sevenfold range of τ, from 0.1 to 0.7, accuracy moves by 0.68 points in total and the five means are mutually overlapping. Realized exposure sits at 1.000 for every run up to $\tau = 0.5$ and 0.999 at $\tau = 0.7$; at $\tau = 0.1$ it exceeds the reference level tenfold and never approaches it. The reference level therefore sets the sign changeover of the exposure term, not a level the controller reaches, and what distinguishes V2 from V1 is that the threshold can decrease at all.

Figure 2 draws both halves of that argument. The left panel is the τ–accuracy curve: it is flat, and the five V2 means lie inside a band 0.68 points wide, so any reading in which the controller steers exposure towards τ predicts a dependence that is not present. The same panel places V1 at $\tau = 1$, which is not an approximation: the V1 rule is exactly the V2 rule with the reference level at the ceiling, so the two coincide there, as we verified by running both controllers on a common trajectory. The panel shows the $+4.8$ point gap that the one-sign change produces against V1 — roughly seven times the entire spread across τ. The right panel shows why. Realized exposure is pinned at the ceiling for every run while the reference level rises from 0.1 to 0.7, so the exposure term stays near saturation and the controller never receives a signal that would pull it back towards τ.

One systematic trend does survive, and we report it rather than smoothing it over: mean clipping intensity falls monotonically with τ, from 0.825 at $\tau = 0.1$ to 0.697 at $\tau = 0.7$. A larger reference level relaxes the tightening term and so clips somewhat less, exactly as the rule dictates. What τ does not do over this range is change the accuracy outcome, which is the property the tracking reading required.

6.1. Diagnostics and negative controls

The A3 ablation shows why exposure feedback was retained during controller development. On the stable FashionMNIST-CNN setting, the original DAGC reaches $79.38 \pm 1.20\%$, compared with $77.63 \pm 1.63\%$ after re-

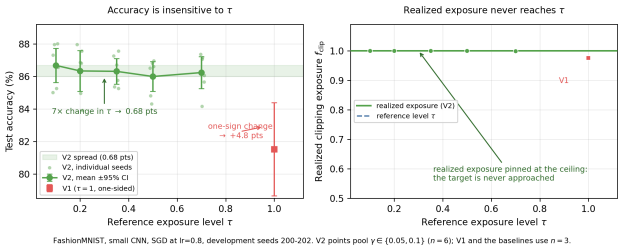


Figure 2. The reference exposure level τ does not control realized exposure. Left: accuracy against τ for V2, with individual seeds, means, and 95% intervals; the shaded band spans the five V2 means. V1 is placed at $\tau = 1$, the value that exactly reproduces the one-sided rule. Right: realized clipping exposure against τ, with the reference level itself shown dashed. Development seeds, FashionMNIST, small CNN, SGD at learning rate 0.8.

Table 2. Development sweep of the reference exposure level τ (FashionMNIST, small CNN, SGD at learning rate 0.8, seeds 200–202, three seeds per cell). Each cell pools the two values of γ tested (0.05 and 0.1), so $n = 6$; accuracy is mean test accuracy with a normal-approximation 95% interval. Realized exposure never falls below 0.999 and is never near any τ tested, while mean clipping intensity decreases monotonically in τ.

τ	n	Accuracy (%)	f_{clip}	i_{clip}
0.10	6	86.68 ± 1.05	1.000	0.825
0.20	6	86.33 ± 1.26	1.000	0.809
0.35	6	86.31 ± 0.79	1.000	0.788
0.50	6	85.99 ± 0.91	1.000	0.758
0.70	6	86.23 ± 0.98	0.999	0.697
V1 ($\tau = 1$)	3	81.52 ± 2.87	0.976	0.435
Fixed (calibrated)	3	84.13 ± 1.05	1.000	0.695
AGC (published)	3	84.58 ± 2.48	1.000	0.926

moving exposure feedback and $77.67 \pm 1.64\%$ with a frozen threshold. Removing alignment feedback gives $78.90 \pm 0.58\%$. More directly, the exposure and frozen variants reduce threshold switches from 8.2 per run to 1.8 and 1.4 while raising clipping intensity from 0.493 to approximately 0.626. This is a behavioural ablation, not evidence of a stable-regime accuracy advantage: its five seeds are small in number and it precedes V2.

The A4 grid supplies the complementary negative control. Across all sixteen dataset-model-optimizer cells at low learning rates, no clipping matches or exceeds the original DAGC. For example, FashionMNIST-CNN with SGD yields $82.27 \pm 1.68\%$ without clipping versus $79.90 \pm 0.81\%$ with DAGC; MNIST-MLP with SGD yields $92.82 \pm 0.56\%$ versus $92.18 \pm 0.55\%$. The absence of a gain where the optimizer is already stable is expected and limits the intended claim to the unstable regime. It also guards against interpreting adaptive clipping as a universal accuracy improvement.

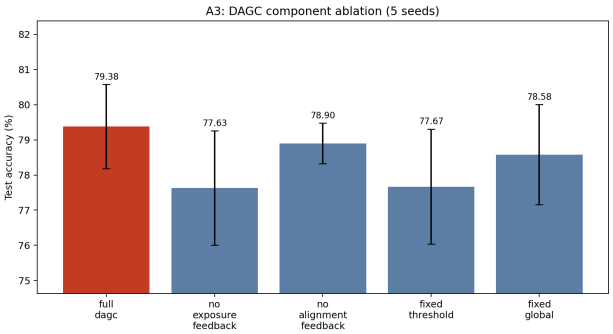


Figure 3. A3 diagnostic ablation of the original controller on FashionMNIST-CNN at a stable learning rate. Error bars summarize five seeds. The experiment motivates exposure feedback but is not used as a V2 performance claim.

Table 3. Mechanism test, eight paired held-out seeds. Accuracy intervals are normal-approximation 95% intervals. I is mean clipping intensity.

Policy	Accuracy (%)	I
Full V2	86.79 ± 0.55	0.798
No alignment	86.63 ± 0.77	0.790
No exposure	81.33 ± 1.03	0.399
Frozen threshold	80.74 ± 0.86	0.346
Intensity-matched fixed	85.61 ± 0.77	0.762
Calibration-trace replay	86.58 ± 0.46	0.789
AutoClip ($p = 0.10$)	79.88 ± 1.44	0.232
Rate tracking	75.21 ± 1.57	0.160

6.2. Discriminating mechanism experiments

We perform a new eight-seed paired experiment on FashionMNIST-CNN at learning rate 0.8, retaining the fifteen-epoch protocol. Calibration seeds 800–802 are disjoint from evaluation seeds 900–907. The calibration V2 traces define an open-loop replay schedule by averaging thresholds at each training step. A fixed threshold is chosen from $\{0.05, 0.10, 0.20, 0.30, 0.50\}$ solely to minimize the discrepancy in calibration mean clipping intensity; this selects 0.20. No evaluation scores select either control.

Comparison with the closest feedback mechanisms. AutoClip uses the tenth percentile of all gradient norms observed through the current step. Rate tracking uses the same initial threshold, exposure EMA, gain, and saturation as V2, but reverses the exposure sign and removes the direction term: $c^+ = c \exp[0.05 \tanh(0.3(E - 0.1))]$. This is a controlled rate-tracking analogue, not the original private federated algorithm. V2 outperforms these specified configurations by 6.92 ± 1.27 and 11.59 ± 1.63 paired percentage points, respectively. These configurations are not comprehensively tuned, so the result distinguishes feedback behaviour in this regime, rather than establishing method-wide superiority.

Table 4. Learning-rate perturbations: eight paired seeds per policy.

Policy	Pulse accuracy (%)	Drop accuracy (%)
Full V2	86.47 ± 0.59	86.88 ± 0.43
No alignment	86.49 ± 0.53	86.66 ± 0.44
Trace replay	86.53 ± 0.81	86.87 ± 0.65

Is a smaller threshold sufficient? V2 exceeds the calibration-intensity-matched fixed rule by 1.18 ± 0.47 paired points, with seven wins in eight seeds. Evaluation intensities are close but not identical (0.798 versus 0.762); this finite-grid control is approximate. Crucially, replaying the calibration threshold trace gives 86.58%, and V2's paired advantage is only 0.21 ± 0.53 points (five wins). Thus a useful time-varying threshold profile reproduces most of the gain without online feedback. The experiment does not identify an additional benefit from reacting to the current evaluation trajectory.

Is the direction signal necessary? Removing alignment changes accuracy by only 0.16 ± 0.41 paired points; V2 wins four comparisons, ties two, and loses two. Removing exposure costs 5.47 ± 1.19 points, with V2 winning all eight comparisons. These results support exposure-driven threshold adaptation in this setting, but do not support necessity of the direction term. An exposure-only controller is a competitive simplification. Neither comparison establishes that reversal predicts future loss deterioration.

6.3. Does online feedback help after a learning-rate change?

We additionally test two prespecified perturbations using independent paired seeds 1000–1007. In the pulse condition, the learning rate changes from 0.8 to 1.6 immediately before zero-based step 100 and returns to 0.8 before step 160. In the drop condition it changes from 0.8 to 0.2 before step 100 and remains there. The 300-step budget, data protocol, momentum buffer, and V2 settings are unchanged. The replay schedule is still the original unperturbed calibration trace; neither condition recalibrates it.

V2's paired advantage over replay is -0.06 ± 0.54 points under the pulse and $+0.01 \pm 0.43$ under the drop. Against the no-alignment variant, its advantage is -0.02 ± 0.43 and $+0.21 \pm 0.35$ points, respectively. Thus neither perturbation identifies additional accuracy benefit from online feedback or the direction term. These results reject the proposed empirical discriminator in these settings: the predetermined schedule remains competitive even after a learning-rate change. They do not show equivalence outside the tested perturbation magnitudes or training budget.

7. Discussion and Limitations

The new controls narrow the mechanism claim. Exposure removal harms performance, but direction removal does not produce a detectable accuracy change. A calibration-derived open-loop schedule nearly matches online V2. The current evidence therefore supports generating a protective threshold profile, not a unique advantage of real-time direction-sensitive feedback. Distinguishing these alternatives is part of the contribution of the mechanism experiments. The evidence supports a narrow claim: under the tested high-learning-rate, small-data protocols, bidirectional exposure feedback improves the clipped training outcome over a separately calibrated fixed threshold and AGC. The paired design, separate fixed-baseline calibration, and frozen V2 setting reduce the most obvious sources of selection bias. The low-learning-rate diagnostic benchmark from the development stage remains a negative control: clipping is not expected to improve accuracy when the unconstrained optimizer is already stable.

One result narrowed our own interpretation. We introduced V2 to track a target exposure, and the τ sweep shows it does not: over a sevenfold range accuracy is flat in τ and realized exposure saturates at the ceiling, even though clipping intensity does respond to τ as the rule dictates. What the correction reliably provides is a threshold that can decrease, and in the unstable regime the controller then holds clipping on for essentially every step. That is a real mechanism, and it explains the V1 failure, but it is not the mechanism we set out to test, and a reviewer should read the V2 gain as evidence for permissive-versus-committed threshold control rather than for target tracking. The dissociation matters for how the hyperparameter should be tuned: τ controls how hard the controller clips, not whether the threshold may fall, so it should be selected for intensity rather than treated as the quantity the controller regulates. A cleaner test of the tracking hypothesis would vary τ together with the learning rate and ask whether a nonzero τ ever binds; our sweep suggests it may not in MNIST-family settings.

The limitations are substantial. The present experiments use MNIST-family data, small CNN/MLP models, short schedules, and normal-approximation intervals with eight seeds. Only fixed global clipping receives per-condition tuning; AGC is reported with its standard coefficient rather than an exhaustive tuning budget, and the same is true of the ZClip and AdaGC baselines, so no claim about those methods is available from these runs. The V2 controller itself is selected on one FashionMNIST-CNN configuration, so its apparent transfer must be verified on larger and more diverse benchmarks. A full submission should add three-channel datasets, larger architectures, longer schedules, and a matched tuning budget for every adaptive baseline.

8. Conclusion

DAGC-V2 adapts a global clipping threshold through bidirectional exposure feedback and consecutive-gradient dynamics. Its defining correction over V1 is simple: making the exposure term change sign allows persistent clipping to tighten the threshold rather than merely slow its relaxation, and in the tested regime that is what separates underclipping from committed clipping. Across four disjoint held-out evaluations, the frozen controller consistently exceeds calibrated fixed clipping and AGC. A sweep of the reference level shows the benefit does not come from tracking a target exposure, and we report it that way. The new mechanism controls show that exposure-only adaptation and a calibration-derived schedule capture most of the gain; they do not establish necessity of directional feedback. These results motivate focused validation, while their MNIST-family diagnostic scope sets clear limits on the claims that can be made at present.

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A. Reproducibility Details

Source code, evaluation scripts, and seed-level experimental artifacts are available at https://github.com/alexsongyzs-dotcom/Dynamics_Aware_Gradient_Clipping. This public manuscript includes the repository link; an anonymous review copy should use an anonymized code archive in accordance with the submission rules.

Table 5. Shared experimental settings. Values not listed are fixed by the policy definition or the calibration protocol.

Setting	Value
Framework / device	PyTorch / CUDA
Optimizer	SGD, momentum 0.9
Weight decay	10^{-4}
Epochs / batch size	15 / 256
Train / test subset	5,000 / 2,000
Projection dimension	8
Divergence loss / jump	100 / $\max(50, 20\ell_{t-1} + 1)$
V2 $(c_0, \gamma, \beta, r, w, \tau)$	$(0.5, 0.05, 0.9, 0.3, 3.0, 0.1)$
AGC coefficient / epsilon	0.01 / 10^{-3}

The evaluation code records complete seed-level outcomes in CSV files. The controller state resets at the beginning of each run. Dataset subsets are sampled deterministically from a seed, and a separate generator controls the shuffled batch order. Consequently, policies with different control trajectories do not accidentally see a different minibatch sequence.

B. Calibration and Held-Out Splits

The V2 development sweep uses seeds 200–202 on FashionMNIST-CNN at learning rate 0.8. It considers exposure targets 0.10, 0.20, and 0.35 and gains $\gamma \in \{0.05, 0.10\}$, alongside fixed clipping, AGC, and V1. The selected V2 configuration in Table 5 is then not changed. Fixed clipping thresholds are selected separately on seeds 400–402 for FashionMNIST-MLP and 600–602 for MNIST-CNN. FashionMNIST-CNN uses an earlier disjoint calibration run. All reported evaluation seeds are absent from these development and calibration splits.

This split does not eliminate all selection effects: the unstable learning-rate band follows an earlier pilot. It does prevent reported evaluation seeds from selecting the V2 target or fixed threshold. This distinction matters because an overly restrictive fixed threshold can make an adaptive policy look better for the wrong reason.

C. Additional Seed-Level Interpretation

Paired differences are evaluated within a seed, not by subtracting unpaired aggregate means. V2 beats calibrated fixed clipping in 6/8, 7/8, 8/8, and 8/8 seeds across the four conditions; it beats AGC in 5/8, 8/8, 8/8, and 8/8. The smaller 5/8 margin at FashionMNIST-CNN with learning rate 0.4 is reported deliberately: it shows the effect is smaller when optimization is less stressed. The result is directional, rather than an assertion that every unstable setting benefits equally.

V1 supplies a useful failure case. It remains stable under clipping but underperforms both fixed clipping and V2 in every held-out condition. This separates avoiding divergence from preserving useful learning dynamics. The bidirectional correction allows the controller to reduce its threshold rather than merely slow the rate at which it increases, and the measured difference in realized clipping intensity between V1 and V2 in each condition is the direct trace of that change.

D. Mechanism Experiment: Complete Protocol

D.1. Data, architecture, and training budget

This experiment uses FashionMNIST, normalized with mean 0.286 and standard deviation 0.353. Each seed selects 5,000 training and 2,000 test images without replacement. Training indices use the run seed; test indices use the seed plus one. The CNN has two 3×3 convolutional layers with 32 and 64 channels, each followed by ReLU and 2×2 max pooling. Its classifier is a flattened $64 \times 7 \times 7$ representation, a 128-unit ReLU layer, and a ten-class output. Cross-entropy is optimized with SGD at constant learning rate 0.8, momentum 0.9, and weight decay 10^{-4}. Batch size is 256; there are 20 minibatches

per epoch and 300 planned steps over 15 epochs. Data-loader workers are set to zero. All policies within a seed share subset, initialization, and shuffled minibatch order.

D.2. Calibration and frozen controls

Calibration uses seeds 800–802. We run V2 and five fixed thresholds on each seed, giving 18 calibration runs. Evaluation uses seeds 900–907 and eight policies, giving 64 independent evaluation runs. The following choices are frozen before evaluation.

Intensity matching. For step t, clipping intensity is $I_t = \max(0, 1 - c_t/(\|g_t\| + 10^{-12}))$; a run's intensity is its mean over completed steps. We average run intensities across the three calibration seeds. The V2 mean is 0.795853. We choose the fixed threshold minimizing the absolute discrepancy with this mean, without using accuracy for selection. The grid and absolute discrepancies are:

Threshold	0.05	0.10	0.20	0.30	0.50
Absolute intensity discrepancy	0.179211	0.123842	0.025591	0.160509	0.383605

This selects $c = 0.20$. Matching is approximate, since the grid is finite and calibration and evaluation seeds differ.

Open-loop replay. For each step t, the predetermined threshold is $\bar{c}_t = \frac{1}{3}\sum_{s=800}^{802} c_{s,t}^{\mathrm{V2}}$. We average the thresholds actually applied to that step, not their next-step values. Evaluation replays this 300-step schedule without observing gradient norms or directions to select the threshold; the current norm is used only to apply clipping. The schedule is constructed solely from calibration runs and is shared by all evaluation seeds.

D.3. Exact policy definitions

Full V2. $c_0 = 0.5$, $\gamma = 0.05$, exposure and alignment EMA decay both 0.9, exposure gain $r = 0.3$, alignment gain $w = 3$, and reference $\tau = 0.1$. The running norm window is 200 steps. Bounds are 0.1 and 10 times the rolling median, with numerical floors 10^{-12} and 10^{-9}. Exposure state starts at zero and alignment state at one.

No alignment. Full V2 with $w = 0$; all other settings remain equal.

No exposure. Full V2 with $r = 0$; alignment and scale bounds remain active.

Frozen threshold. Full V2 with $\gamma = 0$. This disables the multiplicative update, but the running scale projection remains active, so it is not strictly constant. The genuinely constant control is the fixed rule.

Intensity-matched fixed. Constant global threshold $c = 0.20$.

Calibration-trace replay. Apply the predetermined $\bar{c}_t$ above.

AutoClip. At step t, compute the empirical tenth percentile of all raw gradient norms from steps 1 through t, including the current norm, and apply it as the threshold. There is no warmup or moving-window truncation.

Rate tracking. Initial threshold 0.5, exposure EMA initialized at zero with decay 0.9, reference 0.1, and update $c_{t+1} = c_t \exp[0.05 \tanh(0.3(E_t - 0.1))]$ after applying c_t. There is no alignment penalty or rolling-median projection. This is a specified non-private tracking analogue, not a reproduction of DP-FedAvg.

AutoClip and rate tracking use prespecified settings without a matched hyperparameter search. Their results should be interpreted as configuration comparisons. The frozen variant retains scale adaptation through its bounds; its name does not imply that every source of threshold movement is removed.

D.4. Logging, evaluation, and statistical analysis

We store per-step loss, raw gradient norm, clipping coefficient, actual parameter update norm, consecutive full-gradient alignment, exposure, intensity, signed threshold crossing, and applied threshold. Alignment used by the controller is computed from the full flattened gradients; the eight-dimensional fixed random projection is used for additional diagnostics.

A divergent run is flagged for nonfinite loss or gradient norm, loss above 100, or $\ell_t > \max(50, 20\ell_{t-1} + 1)$ for a finite previous loss. Final test accuracy is measured after training and retained for all runs, including flagged runs.

For each alternative b, we compute $d_s = 100(\text{accuracy}_{s,\text{V2}} - \text{accuracy}_{s,b})$ for each of the eight seeds. We report $\bar{d} \pm 1.96 s_d/\sqrt{8}$, count strict positive differences as wins, and count exact equal scores separately as ties. These are descriptive normal-approximation intervals, not multiplicity-adjusted hypothesis tests. No step-level measurements are treated as independent accuracy replicates.

D.5. Reproduction and saved artifacts

From the experimental repository root, run

```
python scripts/run_v2_mechanism.py
```

Outputs are written to `results/v2_mechanism/`. Each calibration and evaluation run has a JSON file containing its summary and complete trajectory. `frozen_protocol.json` records the seeds, intensity matching errors, selected fixed threshold, and replay schedule. `raw_runs.csv` contains the 64 evaluation rows; `summary.csv` contains aggregate and paired statistics. Existing per-run JSON files are reused when resuming. Files must be moved to a fresh output location before changing the protocol, since cached results are not automatically invalidated by configuration edits.

E. Learning-Rate Perturbation Reproduction

Run `python scripts/run_v2_lr_perturbation.py`. The script records the two perturbation schedules and seeds in `results/v2_lr_perturbation/protocol.json`. Each of the 48 runs has a trajectory JSON, and `raw_runs.csv` and `summary.csv` contain final outcomes and paired comparisons. Step indices are zero-based; learning-rate changes occur before the indicated batch's forward pass, with the momentum state retained. The original calibration replay schedule is read from `results/v2_mechanism/frozen_protocol.json`. V2, no alignment, and replay are compared under the same perturbation within each seed. No new controller parameters or replay traces are selected using these evaluations.